Effects of maternal probiotic supplementation on breast milk microbiome and infant gut microbiome and health: a systematic review and meta-analysis of randomized controlled trials

Bekalu Kassie Alemu, MSc; Getnet Gedefaw Azeze, MSc; Ling Wu, MD, PhD; So Ling Lau, MD; Chi Chiu Wang, MD, PhD; Yao Wang, PhD

OBJECTIVE: The early-life microbiome is formed during the perinatal period and is critical for infants' lifelong health. This is established by maternal-infant microbiome crosstalk, which is mediated by the breast milk microbiome. The milk microbiome is dependent on the maternal gut microbiome, suggesting that it could potentially be restored through oral probiotic supplements. Therefore, we conducted this systematic review and meta-analysis to summarize the effect of maternal probiotic supplements on breast milk and infant gut microbiome composition and on infant health.

DATA SOURCES: The PubMed, EMBASE, Web of Science, Scopus, CINAHL, and Science Direct databases were searched until December 15, 2022.

STUDY ELIGIBILITY CRITERIA: Randomized controlled trials following the population, intervention, comparison, and outcome (population: pregnant or lactating women; intervention: probiotics; control: placebo or follow-up; outcome: breast milk and infant gut microbiome composition and infant health) principles were included.

METHODS: Using a random effect model, the standard mean difference, risk difference, and risk ratio with 95% confidence interval were used to measure each outcome. All analyses were conducted using the intention-to-treat approach. Heterogeneity was evaluated using I^2 statistics.

RESULTS: The final data set included 24 randomized controlled trials with a total of 2761 mothers and 1756 infants. The overall effect of probiotics on the beneficial bacteria detection rate in breast milk had a risk difference of 24% (95% confidence interval, 0.1−0.37; $P<.001$; $I^2=91.12\%$). The pooled mean beneficial and pathogenic bacteria abundance in breast milk had a standard mean difference of 1.22 $\log_{10}$ colony forming units/mL (95% confidence interval, 0.48−1.97; $P<.001$; $I^2=95.51\%$) and −1.05 $\log_{10}$ colony forming unites/mL (95% confidence interval, −1.99 to −0.12; $P=.03$; $I^2=96.79\%$), respectively. The overall abundance of beneficial bacteria in the infant gut had a standard mean difference of 0.89 $\log_{10}$ colony forming units/g (95% confidence interval, 0.22−1.56; $P=.01$; $I^2=95.01\%$). It also controlled infant weight gain (standard mean difference, −0.49 kg/equivalent age; 95% confidence interval, −0.82 to −0.17; $P<.001$; $I^2=0.00\%$) and decreased the occurrence of infantile colic (risk ratio, 0.30; 95% confidence interval, 0.16−0.57; $P<.001$; $I^2=0.00\%$).

CONCLUSION: Maternal probiotic supplements effectively orchestrate the breast milk and infant gut microbiome with a wide range of clinical benefits and safety. *Lactobacillus, Bifidobacterium, Streptococcus thermophilus,* and *S. boulardii* can be used as maternal supplements to promote infant health.

Key words: breastfeeding, breast milk microbiome, gut microbiome, maternal-infant health, probiotics, randomized controlled trials, systematic review and meta-analysis

Introduction

The early-life microbiome regulates the development of the cognitive and immune systems during the perinatal period.[1,2] It also determines the occurrence of diseases, including metabolic disorders throughout life.[3] The maternal microbiome is transferred horizontally and vertically through mobile genetic elements during pregnancy to sculpt the infant gut microbiome.[1,4] In addition, accumulating studies have suggested that

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AJOG MFM at a Glance

Why was this study conducted?

This meta-analysis aimed to summarize published randomized controlled trials on the effect of maternal probiotics supplementation during pregnancy and lactation on breast milk and infant gut microbiome composition and on infant health.

Key findings

Probiotic supplements during the perinatal period can (1) effectively shape the breast milk microbiome compositions (improved beneficial bacteria abundance and reduced pathogenic bacteria); (2) remodel the infant gut microbiome profile through breastfeeding; (3) and exert a beneficial influence on the newborn's health including optimizing weight gain and reducing the occurrence of infantile colic.

What does this add to what is known?

Our results confirmed that perinatal oral probiotic supplementation is an impactful intervention during pregnancy and lactation with substantial health benefits for infants.

breast milk acts as an important mediator of microbiome transfer from mother to infant.[5]

Breast milk is a complex biologic system, the bulk of which consists of microorganism communities and other nutritionally valuable elements.[5] It is crucial for early-life microbiome ecology establishment and influences the infant's lifelong health.[6–9] The milk microbiome composition is mainly regulated by entero-mammary and oro-mammary pathways.[10–12] The breast milk microbiome associated with pump feeding has a significant association with environmental bacteria and differs from direct nursing from the breast, indicating the presence of oro-mammary pathways.[10,13,10] Furthermore, a bulk of bacteria in the precolostrum breast milk during pregnancy mimics common infant oral bacteria, suggesting that the gut-breast axis determines the breast milk microbiome.[14] In this gut-breast axis, dendritic cells smuggle beneficial bacteria from the gut to the mammary epithelium, which plays an essential role in programming the mammary system to establish the nutritional, microbiological, immunologic, and neuroendocrine requirements for the newborn.[10,12,15]

Probiotics are live microorganisms that confer health benefits to the host.[16] They are widely supplied to improve gut dysbiosis and to improve luminal and systemic health.[17,18] Literature has shown that probiotic treatments during pregnancy are safe for mothers and their offspring.[19] Some studies suggest altered rates of gestational diabetes, preeclampsia, perinatal mental health disorder, and mastitis.[20–23] However, data on the effect of probiotics on such pregnancy complications remains conflicted.[24–26] Varied conclusions have been drawn from the evidence regarding the effects of probiotics on the maternal and fetal gut microbiome.[27–29] Furthermore, the impact and underlying mechanism of maternal probiotic treatments on newborns' health are still obscure. Given the existence of the gut-breast axis, we reasoned that the breast milk microbiome is a candidate to transfer the probiotic-changed gut microbiome from the mother to the newborn.[12,30]

Objective

To address the current contradictory conclusions and to further evaluate the efficacy of probiotic treatment in the remodeling of breast milk and the infant gut microbiome and infant health outcomes, we conducted this systematic review and meta-analysis of randomized controlled trials (RCTs) with the following specific objectives. First, we assessed the beneficial bacteria detection rate in breast milk. Second, we determined the abundance of beneficial bacteria in breast milk. Third, the levels of pathogenic bacteria in the breast milk were measured. For the fourth, fifth, and sixth objectives, we evaluated the beneficial infant fecal bacteria abundance, infant weight gain, and infantile colic risk reduction effect of the intervention.

Materials and Methods

This systematic review and meta-analysis of RCTs were conducted according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (2020) checklist.[31]

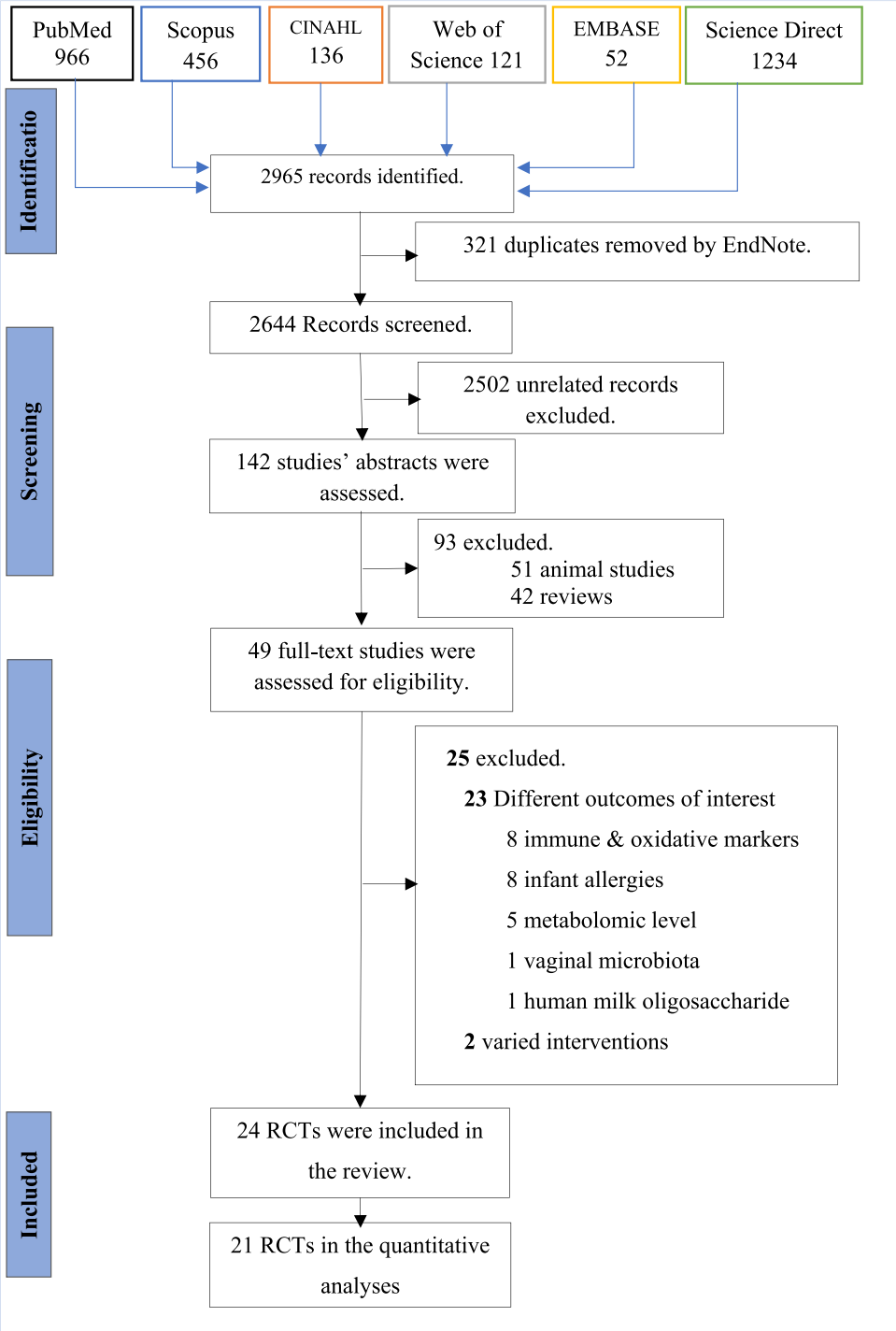
Eligibility criteria

In this review, the population, intervention, comparison, and outcome strategy was followed to select all articles that met the following inclusion criteria. The population was pregnant or lactating mothers and their infants. The intervention was probiotics (either single- or multistrain). The comparator included various controls (placebo, standard treatment, or only follow-up), and the outcomes were breast milk microbiome, infant gut microbiome, infantile colic, and infant weight. Only RCTs, without publication period or place restrictions, were included. Studies carried out at the institution level or community level with individual participant-level randomization that were open, single-blinded, or double-blinded studies were considered, irrespective of the health status of the women and the newborns. Studies with outcomes of interest other than breast milk composition, infant gut microbiome composition, infantile colic, and infant weight were excluded from the study (Figure 1).

Information sources and search strategy

Six electronic databases (PubMed, Embase, Web of Science, Scopus, CINAHL, and Science Direct) were systematically searched for articles published until December 15, 2022. Different search terms were used for each database in 7 main contexts (mothers, probiotics, randomized controlled trials, breast milk microbiome, infant gut microbiome, infant weight

FIGURE 1
Selection of randomized controlled trials using a PRISMA flow chart



PRISMA, Preferred Reporting Items of Systematic Reviews and Meta-Analyses; *RCTs*, randomized controlled trials.

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gain, and infantile colic). Details of the search strategies are available in the Supplemental Material (Supplemental Table S1) for all the databases searched. Moreover, to obtain further articles, reference lists and additional files of the included studies and related systematic reviews were retrieved.

Study selection and data extraction

Two independent authors (B.K.A. and G.G.A.) screened each article eligible for the review and extracted effect-

measuring data using a standardized format that comprised the following components: study identifier, first author, year, country, population, intervention, comparator, intervention initiation time, duration of intervention, number of participants in the intervention and control groups, number of participants with and without the outcome, mean beneficial and pathogenic bacteria abundance in breast milk, mean infant fecal beneficial bacteria abundance, mean body weight of infants, number of infants with colic in both groups, bacteria detection method, and side effects. A third reviewer (Y.W.) resolved discrepancies between the 2 reviewers. The primary outcome measures in this review were the probiotic species detection rate and the mean abundance of beneficial bacteria in breast milk. The detection rate was computed from the count data extracted from the included articles. Pathogenic bacterial abundance in breast milk, beneficial bacterial abundance in infant stool, infant body weight, and infantile colic were the secondary outcome measures. We further evaluated the adverse effects of probiotics on fetomaternal health, such as mastitis-associated breast pain, as our additional outcome. However, this outcome was qualitatively described because no adverse effects were reported in most studies included in our meta-analysis. After data extraction, the median with interquartile range (IQR) values of bacterial abundance in breast milk and infant stool were transformed to means and standard deviations (SDs).[32] Data were also extracted from box plots and other graphical presentations using the online software Web-Plot-Digitizer (developed by Ankit Rohatgi in South Bend, Indiana).[33–35]

Statistical analysis

A random-effects model was used for all summary measures in this meta-analysis. In the case of empty cells, a value of 0.5 was added to each cell of specific articles to diminish the zero count in the standardized relative risk (RR) and standard mean difference (SMD) analyses.[36] The effect size measurement of the intervention on beneficial bacteria detection rate in breast milk was expressed using risk difference (RD), whereas mean bacterial abundance in breast milk and infant gut was expressed using SMDs with a 95% confidence interval (CI). We used the Hedges g in estimating the SMD to avoid noneligible bias caused by the small sample size of the studies included.[37] Similarly, its effect on infant body weight was measured using SMDs with 95% CIs. The units used to measure the mean bacterial abundance in breast milk and stool samples were colony forming units per mL (CFU/mL) and CFU/g, respectively. The infantile colic occurrence was expressed as the RR with a 95% CI.

A forest plot was used to observe the overall pooled results and the effect of each individual study. Heterogeneity across studies was assessed using I^2 test statistics. A P value of <.05 was used to indicate heterogeneity and statistical significance. Publication bias was assessed using the asymmetry of funnel plots and Egger's test. Trim and fill analyses were performed for findings with an Egger's test P value of <.5. Subgroup analyses were conducted to manage heterogeneity. To identify the source of heterogeneity and to measure the degree of influence of the individual studies included, meta-regression and sensitivity analyses were performed. The effects of probiotics on genus-specific categories were evaluated for all pathogenic bacterial taxa reported in the included studies. To overcome the multiplicity of effect size in pooling the overall effect of probiotics on different species and genus groups of pathogenic bacteria, random selection of single bacterial taxa using an Excel sheet was performed.[38,39] In addition, a separate genus-label analysis was also performed (Supplemental Figure S6).

Assessment of risk of bias

Two authors (B.K.A. and G.G.A.) independently assessed and evaluated the methodological quality of the included studies. The Risk of Bias 2 (RoB-2) tool was used to assess the included studies according to the 5 domains.[40] Potential confounders were discussed before assessment and result differences in each domain were managed with discussion. One open-label study[41] was determined to be at high risk of bias, and only 2 studies were regarded as having some concerns,[42,43] otherwise, all remaining 21 studies were determined to be low risk (Supplemental Figure S1).

Results

Study selection

In this systematic review and meta-analysis of RCTs, we identified a total of 2965 articles. After 321 duplicate records were removed, we screened 2644 articles. In the final screening, 49 articles were screened by reviewing the full texts of which 25 were excluded by the eligibility criteria (Supplemental Table S2), and 24 articles were included in the systematic review[27–29,41–61] (Figure 1) and 21 in the meta-analysis.

Study characteristics

Across 24 RCTs included in this review, a total of 2761 mothers and 1756 infants were examined. The included studies were published between 2008 and 2022. The majority of the included studies were from Europe (7 from Spain, 4 from Finland, 2 from Norway, 2 from Italy, and 1 from Sweden). There were 2 from Australia, and the rest were from Indonesia, Iran, Korea, Turkey, Japan, and Tanzania (1 each) (Table 1). Overall, 45 bacterial taxa report records[27,28,41–52,54,55] (12 reports for beneficial bacteria detection rate, 10 reports for beneficial bacteria abundance in breast milk (BM), 17 reports for pathogenic bacteria abundance in BM, and 6 reports for beneficial bacteria abundance in infant stool) were included in this meta-analysis. Although we found some reports of *Bifidobacterium* and other beneficial bacteria in terms of detection rate and mean abundance in BM and infant stool, we used reports of *Lactobacillus*, which is a commonly reported bacterial taxon across studies included in the analysis. We categorized the pathogenic bacteria taxa into *Staphylococcus*, *Streptococcus*, aerobes, and anaerobes.

Quality assessment

Using the RoB-2 tool, study quality was assessed across the 5 measurement domains.[40] One open-label study[41] was

TABLE 1
Characteristics of the included randomized controlled trial studies

Trial, country	Subjects	Status	Wk	Intervention	Control	Dose and frequency	Side effects
Jiménez et al,[48] 2008, Spain	Lactating women	With illness	4	*Lactobacillus salivarius* and *L. gasseri*	Placebo	1 capsule/d	No report
Abrahamsson et al,[46] 2009, Sweden	Pregnant women	Healthy	4	*L. reuteri*	Placebo	1 capsule per d	No report
Lahtinen et al,[29] 2009, Australia	Pregnant women	Healthy	4	*L. rhamnosus* GG	Placebo	5 oil droplets/d	No report
Arroyo et al,[47] 2010, Spain	Lactating women	With illness	3	*L. fermentum or L. salivarius*[a]	Antibiotics	1 capsule/d	No report
Leila et al,[27] 2011, Iran	Lactating women	Healthy	4	Probiotic capsule[b]	Follow-up	1 capsule/d	No report
Bisanz et al,[41] 2015, Tanzania	Pregnant women	Healthy	32	*L. rhamnosus* GR-1	Follow-up	250 g of yogurt 6 d/wk	Safe or no SE
Maldonado-Lobón et al,[49] 2015, Spain	Lactating women	With illness	3	*L. fermentum*	Placebo	1 pill TID	No report
Mastromarino et al,[44] 2015, Italy	Pregnant women	Healthy	8	VSL#3	Placebo	1 bag/d	Safe and no SE
Fernández et al,[51] 2015, Spain	Pregnant women	Healthy	8	*L. salivarius*	Placebo	1 capsule/d	Safe and no SE
Dewanto et al,[28] 2017, Indonesia	Pregnant women	Healthy	20	*Bifidobacterium animalis*	Placebo	1 capsule/d	No report
Hurtado et al,[45] 2017, Spain	Lactating women	With illness	16	*L. fermentum*	Placebo	109 unit/d	No report
Simpson et al,[42] 2018, Norway	Pregnant women	Healthy	16	Probiotic milk[c]	Placebo	250 mL probiotic milk/d	No report
Pastor et al,[50] 2020, Spain	Lactating women	Healthy	16	*L. fermentum*	Placebo	1 capsule/d	No report
Shin et al,[43] 2021, Korea	Lactating women	Healthy	8	*L. rhamnosus Saccharomyces boulardii*	Follow-up	1 capsule BID	No report
Grzeskowiak et al,[52] 2012, Finland	Pregnant women	Healthy	16	LPR + *B. longum*	Placebo	One sachet of 7 g/d	No report
Rautava et al,[53] 2012, Spain	Pregnant women	Healthy	2	*B. lactis, B. lactis* + LGG	Placebo	One sachet of powder	Safe and no SE
Dotterud et al,[54] 2015, Norway	Pregnant women	Healthy	22	LGG + *L. acido* + *B. animalis* lactis	Placebo	250 mL of milk	No report
Enomoto et al,[55] 2014, Japan	Pregnant women	With illness	4	*B. breve* + *B. longum*	Placebo	One sachet daily	Safe and no SE
Ismail et al,[56] 2012, Australia	Pregnant women	With illness	4	LGG	Placebo	Once daily	No report
Luoto et al,[59] 2010, Finland	Pregnant women	Atopic history	28	LGG and *B. lactis*	placebo	One capsule/d	No report
Vähämiko et al,[58] 2019, Finland	Pregnant women	With illness	>24	LGG and *B. lactis*	placebo	One capsule/d	No report
Luoto et al,[57] 2010, Finland	Pregnant women	Atopic history	>24	LGG	placebo	One capsule/d	Safe and no SE
Baldassarre et al,[60] 2016, Italy	Pregnant women	healthy	8	Vivomixx/visbiome	placebo	1 sachet/d	Safe and no SE
Karaahmet et al,[61] 2022, Turkey	Lactating women	NR	2	*B. actiregularis*	follow-up	A probiotic drink/d	No report

BID, 2 times per day; *NR*, no reported data but detected; *TID*, 3 times per day.

[a] Two different species of the *Lactobacillus* genus was administered and therefore the sum of detection of the sum of sample size was used.; [b] Interventions used were *Lactobacillus casei, Lactobacillus rhamnosus, Lactobacillus acidophilus, Bifidobacterium longum, Bifidibacterium breve.*; [c] Intervention species were (LGG) *Lactobacillus rhamnosus* GG, *Lactobacillus acidophilus* La-5, *and Bifidobacterium animalis* ssp. *Lactis Bb-12.* VSL#3 or vivomixx contains 8 strains *(Bifidobacterium* and *Lactobacillus)* and the commonly reported *Lactobacillus* genus included in the meta-analysis.

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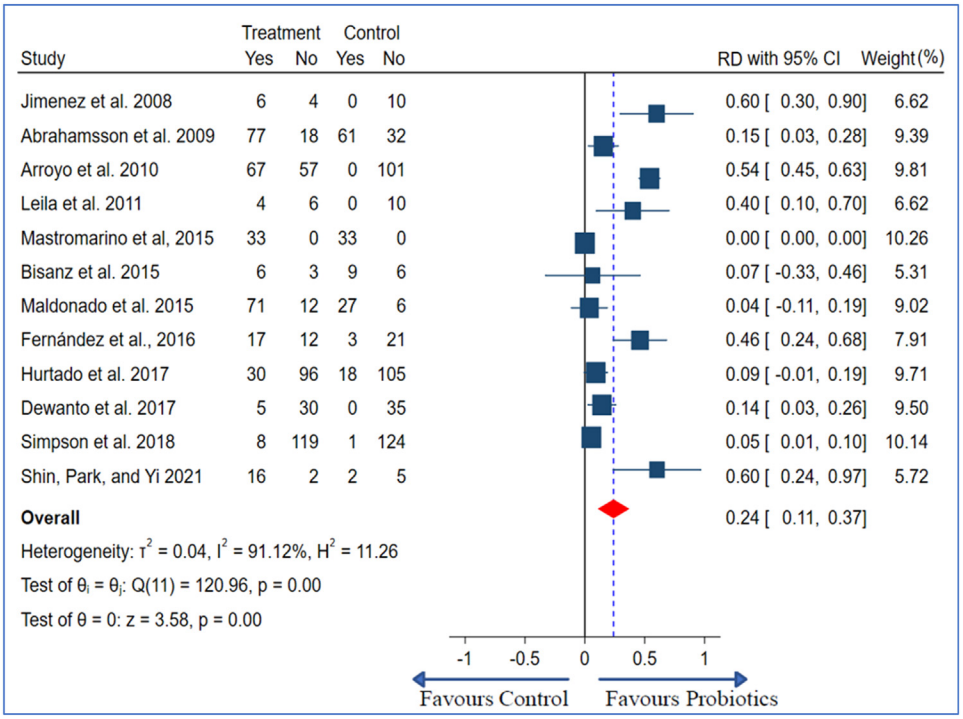
FIGURE 2
Probiotics on beneficial bacteria detection rate in breast milk



CI, confidence interval; *RD*, risk difference.

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determined to be at high risk for bias, and 2 studies were regarded as having some concerns;[42,43] the remaining 21 studies were evaluated to be at low risk of bias (Supplemental Figure S1). Asymmetry of the funnel plot upon visual inspection (Supplemental Figure S2) and the Egger test were used to assess the publication bias.

Synthesis of results

Effect of probiotic supplements on milk microbiome compositions. In breast milk, a greater proportion of beneficial bacteria was found in the probiotic-supplemented group than in the control group. The overall beneficial bacteria detection rate in breast milk was 24% higher in the probiotics group than in the controls (RD, 0.24; 95% CI, 0.11−0.37; $P<.001$; $I^2=91.12\%$) (Figure 2). The asymmetry of the funnel plot (Supplemental Figure S2, A) and the Egger test ($P<.001$) indicated the existence of publication bias in this study. To estimate the number of missing studies that might exist owing to publication bias in the funnel plot and to adjust for the overall effect estimate, trim and fill analysis, which is a nonparametric method, was considered, and 5 studies were identified (Supplemental Figure S3, A). Moreover, subgroup analyses were also conducted for different factors, such as the health status of participants, time of initiation of the intervention, length of duration of the intervention, and region of the study (Europe and non-Europe) (Supplemental Figure S4, A−D). A meta-regression analysis based on the intervention initiation time ($P=.60$), region ($P=.72$), health status of participants ($P=.91$), and duration of intervention ($P=.29$) did not reveal the origin of heterogeneity (Supplemental Table S3, A). The results of the sensitivity analysis showed that there was no outlier study that had an impact on the overall result estimation (Supplemental Figure S9, A), and we retained the result as it was.

To measure the effect of probiotics on the mean abundance of beneficial bacteria in breast milk, we combined 10 studies' reports (Table 2). The overall SMD of beneficial bacteria abundance in breast milk was 1.22 $\log_{10}$ CFU/mL with a 95% CI of 0.48 to 1.97 ($P=.00$; $I2=95.51$) (Figure 3, A). Asymmetry of the funnel plot (Supplemental Figure S2, B) and Egger's test ($P=.03$) demonstrated the presence of publication bias. Trim and fill analysis was conducted, leading to the identification of 5 conceivable studies (Supplemental Figure S3, B). Subgroup analysis of the mean effect size by the duration of the intervention, health status of the participants, time of initiation of the intervention, and study region showed no substantial difference in heterogeneity management, and it implied that there was a significant effect among groups who took the probiotic for more than 4 weeks (SMD, 1.35 $\log_{10}$ CFU/mL; 95% CI, 0.38−2.31), healthy groups of participants (SMD, 1.48 $\log_{10}$ CFU/

TABLE 2
The effect of perinatal probiotics on the detection rate of beneficial bacteria and mean beneficial bacteria abundance in breast milk in log_{10}CFU/mL

Trial	BB detected in BM of PG (n/N)	BB detected in BM of CG (n/N)	Detection method	BB load in BM of PG mean±SD	BB load in BM of CG mean±SD
Jiménez et al,[48] 2008	6/10	0/10	PCR	2.55±0.10	0.±0.70
Abrahamsson et al,[46] 2009	77/95	61/93	NR	NR	NR
Lahtinen et al,[29] 2009	—/59	—/59	qPCR	ND	ND
Arroyo et al,[47] 2010	135/251	0/101	PCR	NR	ND
Leila et al,[27] 2011	4/10	0/10	16s rDNA	1.27±0.96	0
Bisanz et al,[41] 2015	6/9	9/15	16s rRNA	4.6±1.6	3.12±1.17
Maldonado-Lobón et al,[49] 2015	71/83	27/33	qPCR	2.01±0.44	2.23±0.04
Mastromarino et al,[44] 2015	33/33	33/33	RT PCR	3.86±1.74	3.2±0.99
Fernández et al,[51] 2016	17/29	3/24	PFGE	2.85±0.07	2.65±0.04
Dewanto et al,[28] 2017	5/35	0/35	RT PCR	1.71±0.11	0
Hurtado et al,[45] 2017	30/126	18/123	qPCR	3.05±0.64	2.93±0.68
Simpson et al,[42] 2018	8/127	1/125	RT PCT & 16S rRNA	NR	NR
Pastor et al,[50] 2020	—/139	—/152	qPCR	3.3±0.7	3.1±0.40
Shin et al,[43] 2021	16/18	2/18	qPCR	2.48±0.00	1.39±0.00

BB, beneficial bacteria; *BM*, breast milk; *CFU*, colony forming units; *CG*, control groups; *N*, the number of participants in the final analysis of probiotic groups and control groups; *n*, mothers among whom beneficial bacteria were detected in their breast milk in probiotic and control groups; *NR*, no reported data but detected; *PCR*, polymerase chain reaction; *PFGE*, pulsed-field gel electrophoresis; *PG*, probiotics group; *qPCR*, quantitative PCR; rRNA, ribosomal RNA; *RT PCR*, reverse transcriptase PCR; *SD*, standard deviation.

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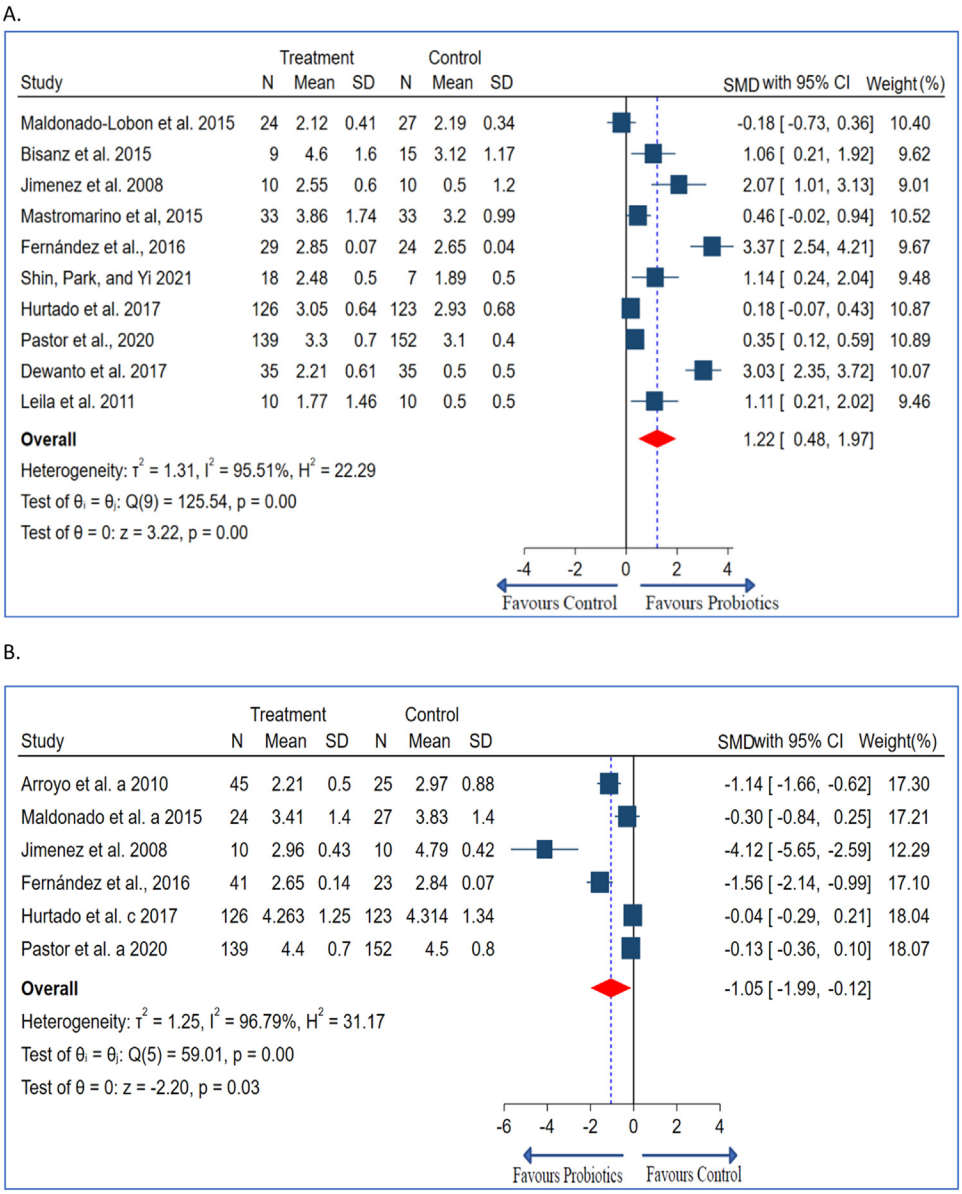
mL; 95% CI, 0.59−2.38), and those who started the intervention during pregnancy (SMD, 1.98 log_{10} CFU/mL; 95% CI, 0.59−3.38) (Supplemental Figure S5, A−C). It was also significant in studies conducted in both European and non-European settings (Supplemental Figure S5, D). The meta-regression for the aforementioned variables did not find the source of heterogeneity among variables, such as region (*P*=.77) and health status of participants (*P*=.52) (Supplemental Table S3, B). Based on the sensitivity analysis (Supplemental Figure S9, B), there were no outlier studies that could impact the pooled estimate.

Approximately 17 pathogenic bacterial taxa were identified in the included studies (Table 3). The pooled suppressive effect of maternal oral probiotic supplementation on pathogenic bacteria abundance in breast milk was measured by the standard mean difference of pathogenic bacteria among mothers who took probiotics and their control comparators. The pooled SMD of pathogenic bacteria in log_{10} CFU/mL was −1.05 with a 95% CI of −1.99 to −0.12 (*P*=.03). Considerable heterogeneity was also observed (I^2=96.79%) (Figure 3, B). The funnel plot of symmetry (Supplemental Figure S2, C) and Egger's test (*P*=.01) showed the existence of publication bias. The trim and fill analysis led to no added study (Supplemental Figure S3, C); therefore, we performed a subgroup analysis by genus category, which showed that probiotics had a significant suppressive effect on *Staphylococcus* (−1.14; 95% CI, −2.15 to −0.13) (Supplemental Figure S6 and S7) with a substantial reduction in heterogeneity among aerobes and anaerobes. The sensitivity analysis showed that no outlier study was detected (Supplemental Figure S8, C). In addition, oral administration of probiotics also has an important effect in managing clinical symptoms of maternal illnesses such as mastitis and breast pain.[45,47−49] Moreover, as our third outcome, no adverse effects were detected during probiotic interventions in either mothers or offspring, indicating that probiotic supplements can be considered as a safe intervention.[41,44,51,53,55,59]

Effect of maternal probiotic supplements on infants' gut microbiome and health. The effect of maternal oral probiotic administration on infant gut microbiome through breastfeeding (Table 4) and its role in reducing infant weight and infantile colic risk are summarized in this review (Table 5).

The mean probiotic bacterial abundance in the stools of infants whose mothers received probiotic supplements during pregnancy and breastfeeding was 0.89 log_{10} CFU/g (95% CI, 0.22−1.56; *P*=.01) more than the controls (Figure 4, A). The symmetry of the funnel plot (Supplemental Figure S2, D) and Egger's test (*P*=.34) revealed no significant publication bias. Therefore, trim and fill analysis was not required. Because it had considerable heterogeneity, we performed

FIGURE 3
Forest plot of mean bacterial load in breast milk



Data are presented as $\log_{10}$ CFU/mL for (**A**) beneficial bacteria and (**B**) pathogenic bacteria.

CFU, colony forming units; *CI*, confidence interval; *SD*, standard deviation; *SMD*, standard mean difference.

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a subgroup analysis by duration of intervention and found no significant reduction in heterogeneity (Supplemental Figure S8). Because we did not have enough categories of variables, we could not perform a meta-regression, but we performed a sensitivity analysis, and no study had an extreme effect (Supplemental Figure S9, D).

We also recapitulated the role of prenatal probiotics in controlling childhood obesity and DNA methylation of obesity-related genes (Table 5). Infants in the control group had about 490 g excess weight gain than those in the probiotic group at the sixth month after birth (SMD, −0.49; 95% CI, −0.82 to −0.17 kg/equivalent age; I^2=0.00%; *P*=.00) (Figure 4, B), whereas the infants in both groups were breast-milk fed. In addition, probiotic interventions reduced the occurrence of infantile colic (RR, 0.30; 95% CI, 0.16−0.57; *P*=.00) (Figure 4, C). The symmetry of funnel plots (Supplemental Figure S2, E) and Egger's test (*P*=.573) showed no significant publication bias. The test of

TABLE 3
Effect of maternal probiotics on mean pathogenic bacterial load in breast milk in log_{10} CFU/mL

Trial	Reported bacteria taxa	Genus category	Milk samples (n) in PG	Mean±SD PB load in PG	Milk samples (n) in CG	Mean±SD PB load in CG
Jiménez et al,[48] 2008	Staphylococci	*Staphylococcus*	10	2.96±0.55	10	4.79±0.23
Arroyo et al,[47] 2010	*Staphylococcus aureus*	*Staphylococcus*	45	2.21±0.50	25	2.97±0.88
Arroyo et al,[47] 2010	*Staphylococcus epidermidis*	*Staphylococcus*	95	2.62±0.49	76	3.31±0.82
Arroyo et al,[47] 2010	*Streptococcus mitis*	*Streptococcus*	32	2.35±0.37	31	3.14±0.72
Arroyo et al,[47] 2010	*Streptococcus salivarius*	*Streptococcus*	3	2.23±0.60	3	3.12±1.09
Arroyo et al,[47] 2010	*Rothia* spp.	Aerobes	7	2.04±0.24	5	2.42±0.67
Maldonado-Lobón et al,[49] 2015	*Staphylococcus*	*Staphylococcus*	24	3.41±1.40	27	3.83±1.40
Maldonado-Lobón et al,[49] 2015	Streptococcus	*Streptococcus*	24	3.03±0.82	27	3.31±0.86
Fernández et al,[51] 2016	Staphylococcal /*Streptococcus*	*Staphylococcus* /*Streptococcus*	41	2.65±0.14	23	2.84±0.07
Hurtado et al,[45] 2017	*Staphylococcus*	*Staphylococcus*	126	3.98±0.69	123	4.26±0.72
Hurtado et al,[45] 2017	*Streptococcus*	*Streptococcus*	126	4.77±0.65	123	4.95±0.75
Hurtado et al,[45] 2017	Aerobes	Aerobes	126	4.26±1.25	123	4.31±1.34
Hurtado et al,[45] 2017	Anaerobes	Anaerobes	126	4.05±0.97	123	4.20±1.25
Pastor et al,[50] 2020	*Staphylococcus*	*Staphylococcus*	139	4.40±0.70	152	4.50±0.80
Pastor et al,[50] 2020	*Streptococcus*	*Streptococcus*	139	5.30±1.00	152	5.10±0.80
Pastor et al,[50] 2020	Aerobes	Aerobes	139	4.10±1.00	152	4.00±1.20
Pastor et al,[50] 2020	Anaerobes	Anaerobes	139	4.20±1.10	152	4.10±1.10

CFU, colony forming units; *CG*, control group; *n*, number of milk samples from which the specified bacterial taxa were detected; *PG*, probiotic group; *SD*, standard deviation.

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TABLE 4
Effect of maternal probiotics on mean beneficial bacterial load in infant stool

Trial	(n) PG	(n) CG	Detection methods	Infant stool BB load mean±SD in PG	Infant stool BB load mean±SD in CG
Mastromarino et al,[44] 2015	33	33	RT PCR	5.01±0.87	4.2±0.99
Lahtinen et al,[29] 2009	59	57	qPCR	NR	NR
Abrahamsson et al,[46] 2009	51	49	NR	5.2±0.6	3.3±0.9
Dewanto et al,[28] 2017	35	35	RT PCR	No effect	No effect
Pastor et al,[50] 2020	139	152	qPCR	7.8±0.9	7.8±1.0
Grzeskowiak et al,[52] 2012	28	22	FCM-FISH & qPCR	1.45±0.56	1.287±0.45
Rautava et al,[53] 2012	20	9	qPCR	NR	NR
Dotterud et al,[54] 2015	122	121	qPCR	3.04±1.25	2.11±1.26
Enomoto et al,[55] 2014	112	32	PCR	1.5±0.56	1.46±0.51
Ismail et al,[56] 2012	50	48	PCR and 16s rRNA	No effect	No effect

BB, beneficial bacteria; *CG*, control group; *FCM-FISH*, Flow cytometry - fluorescent in situ hybridization; *(n) PG* and *n*, number of infants' stool samples used to determine beneficial bacteria load after breastfeeding of probiotic supplemented and control groups, respectively; *NR*, no reported data but detected; *PCR*, polymerase chain reaction; *PG*, probiotic group; *qPCR*, quantitative PCR; *rRNA*, ribosomal RNA; *RT PCR*, reverse transcriptase PCR.

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TABLE 5
Effect of maternal perinatal probiotics administration on (A) infant weight control (B) infantile colic reduction

A: Studies	No. PG	No. CG	Infant Wt. PG	Infant Wt. CG	Remarks
Luoto et al,[57] 2010	67	63	4.259±0.159	4.341±0.192	Probiotics can address the obesity epidemic concerns
Vähämiko et al,[58] 2019	7	8	7.973±1.013	8.733±0.947	Has influence on DNA methylation of obesity-related genes
Luoto et al,[59] 2010	54	59	1.1973±0.251	12.08±0.25	POPS modifies the growth pattern limiting factor
B: Studies	No. PG	No. CG	Develop colic (n) in PG	Developed colic (n) in CG	Remarks
Baldassarre et al,[60] 2016	33	33	3	13	Improves GI symptoms
Pastor et al,[50] 2020	139	152	4	8	Influences infant health
Karaahmet et al,[61] 2022	15	16	3	14	Decrease crying frequency and intensity

CG, control group; *No. PG*, number of infants fed with breast milk from mothers receiving probiotics; *No. CG*, number of infants fed with breast milk from mothers in the control group; *PG*, probiotic group; *POPS*, perinatal oral probiotics supplementation; *Wt*, weight.

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heterogeneity (I^2=0.00%) showed no substantial effect; therefore, a trim and fill analysis and sensitivity analysis was not conducted.

Comment

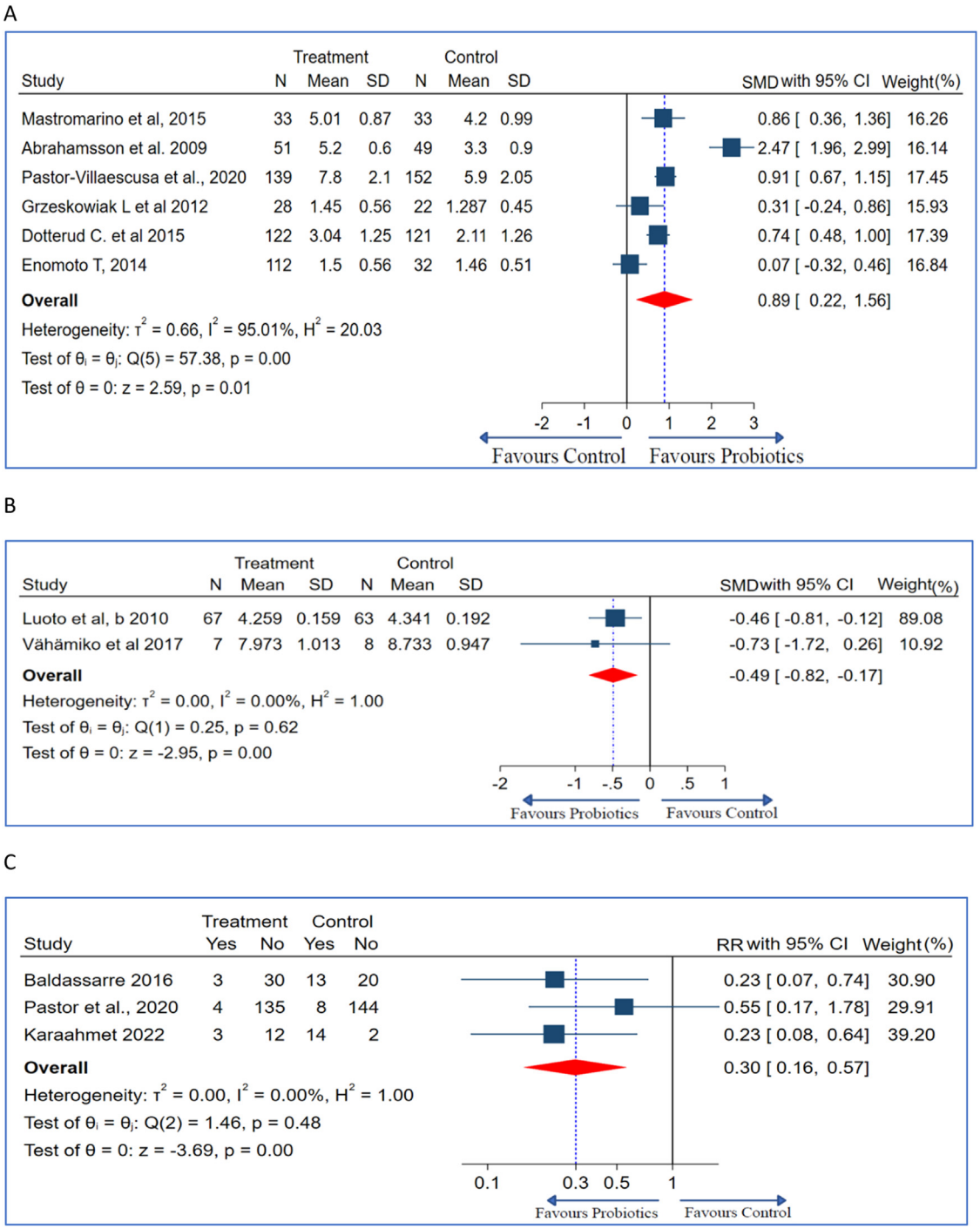
Main findings

This was a systematic review and meta-analysis of RCTs that summarized the effects of maternal oral probiotic administration on the breast milk microbiome, beneficial bacteria composition in the infant's gut, and infant health. Overall, the pooled effect size estimate analysis demonstrated that maternal probiotic supplementation significantly (1) augmented the detection rate of beneficial bacteria; (2) improved the abundance of beneficial bacteria; and (3) suppressed the growth of pathogenic bacteria in breast milk from the maternal side, and this signifies that the milk microbiome is an important mediator in orchestrating maternal-infant crosstalk. Notably, perinatal probiotic supplementation also has a significant impact on (4) improving beneficial infant fecal bacterial abundance, (5) infant weight reduction, and (6) infantile colic reduction.

Comparison with existing literature

The origin of human milk bacteria is assumed to be via the entero-mammary pathway as a route for bacteria to translocate from the maternal gut.[62] Our results are in agreement with the importance of the entero-mammary pathway in the microbial composition of breast milk.[10,14] In our findings, except for 1

FIGURE 4
Effect of probiotics on infant health



A, Infant stool beneficial bacteria abundance. **B,** Infant weight. **C,** Infantile colic risk.

CI, confidence interval; *RR*, relative risk; *SD*, standard deviation; *SMD*, standard mean difference.

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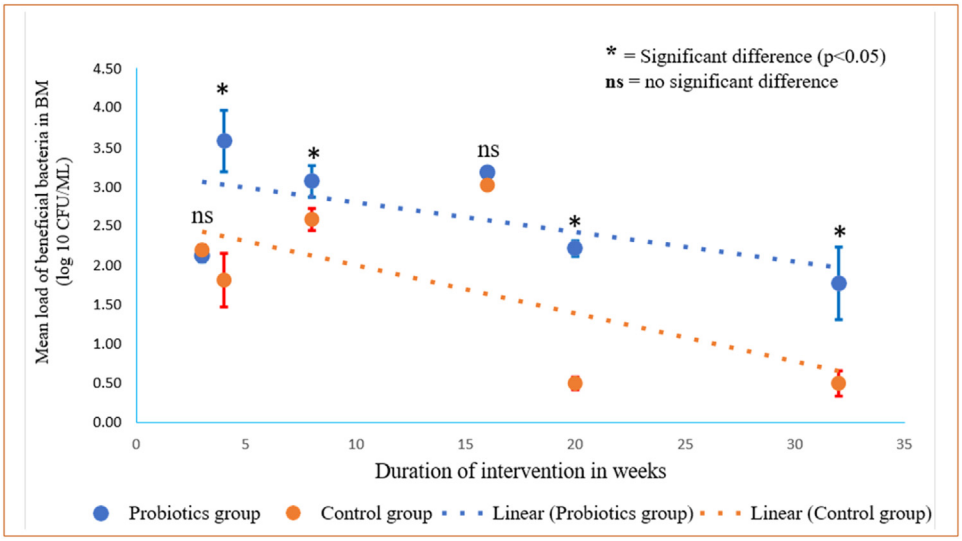
study that showed an equal detection rate,[44] all included studies showed that beneficial bacteria were detected in a greater number of participants in the probiotic groups than in the control groups.[27,28,41–43,46–49,51] Moreover, the overall mean beneficial bacterial abundance in the breast milk of the intervention group was higher than that of the control group. Except for 1 study with the lowest intervention duration by Maldonado et al,[49] studies included

FIGURE 5
Mean beneficial bacterial load in breast milk among probiotic-supplemented and control groups over the duration of the intervention



CFU, colony forming units.

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in the analysis showed higher mean bacterial content in the probiotic groups than in the controls (Figure 5).[27,28,41,43–45,48,50,51] We also found that the pooled SMD of pathogenic bacteria in breast milk was significantly lower among the treatment groups than among the controls. Therefore, our results showed that orally administered probiotics effectively shaped the microbiome composition of breast milk.

Lactobacillus and *Bifidobacterium* are the most common probiotic bacteria used in clinical practice with a wide range of clinical benefits from the management of infection to metabolic disease.[63–66] They remodel the innate and adaptive immunity of the consumer by modulating immune cells such as macrophages and dendritic cells.[67] Furthermore, they suppress pathogenic bacteria in the intestine by limiting adhesion sites and receptors, competing for nutrients, and producing acids that can inhibit pathogens.[68] In our meta-analysis, species in the genera *Lactobacillus (L. salivarius, L. gasseri, L. reuteri, L. rhamnosus* GG, *L. fermentum, L. acidophilus, L. rhamnosus* GR, *L. fermentum, L. plantarum, L. paracasei L. delbrueckii)* and *Bifidobacterium (B. animalis, B. longum, B. breve, B. lactis, B. actiregularis, B. infantis)* were probiotics administered to pregnant and lactating mothers in different formulations. In addition, the species *Saccharomyces boulardii* and *Streptococcus thermophilus* were considered probiotic species from the genera *Saccharomyces and Streptococcus*.[27–29,41–61] Of note, none of the included studies reported any adverse events in the mothers or offspring during the intervention, suggesting that probiotic supplement is a safe treatment during pregnancy and the postpartum period.[19,41,44,51,53,55,59,69]

Probiotics have the effects of promoting the abundance of beneficial bacteria, boosting the immune system, maintaining the luminal surface intact, and modulating gut dysbiosis associated with gastrointestinal diseases.[46] Probiotic supplementation during pregnancy and lactation has positive effects on the mothers' health, and there is evidence to prove that probiotics reduce breast pain[47,49] and have therapeutic and preventive effects on mastitis and its associated symptoms.[23,45,48,70]

Breastfeeding is regarded as an ideal practice for babies' nutrition because breast milk is a ready-made, demand-based, species-specific, and easily digestible substance for newborns.[71] It contributes to the physiological immune system programming in newborns by translocating cytokines, maternal antibodies, and T-cells from the mother.[72] It also benefits cognitive and intellectual capacity development, psycho-affective maturation, and mother-infant bonding.[73]. Our meta-analysis findings provide additional evidence on the benefits of breastfeeding through the merit of early-life gut microbiome establishment. This finding potentiates the role of maternal probiotic supplementation in infant health. Probiotics partially promote infant health by enhancing the beneficial bacterial load of breast milk. Our results demonstrate that perinatal maternal probiotic supplementation modulates the gut microbiome of breastfed infants by improving the abundance of beneficial bacteria.[46,52–56] It also reduces infant weight in the first year of life.[57–59] Even though a concrete weight gain limit is not set, a study by Vähämiko et al[58] stated that the weight gain and reduction in the risk of obesity may be associated with the DNA

methylation status of certain promoters of obesity and weight gain-related genes both in women and their infants. Furthermore, it is also recognized that maternal oral probiotic supplementation could reduce the incidence rate ratio of infantile colic. This is supported by another systematic review that revealed that probiotics have a colic reduction effect in exclusively breastfed infants.[74]

Strengths and limitations

Our meta-analysis summarized RCTs that assessed the effect of oral maternal probiotic supplementation on the breast milk and infant gut microbiome and breastfed infants' health and had its own strengths. First, we followed a comprehensive search strategy, study selection criteria, data extraction strategy, and risk of bias assessment using standard tools and 2 independent reviewers.[40] The overall risk of bias assessment showed that 95.8% of studies had a low risk of bias, and 4.2% of studies had some concerns of bias (Supplemental Figure S1). Second, our findings were based on a considerable sample size (2761 mothers and 1756 infants). In general, our results may attract clinicians' and researchers' interest in the importance of the human milk microbiome in maternal-infant crosstalk and in formulating its composition by probiotic interventions.

Our study has limitations, such as the search for records was limited to reports published in electronic databases. In addition, there were also uncontrollable heterogeneities in the subgroup analysis in the meta-analysis, which could be because of the huge variation in sample size from 9 to 251 participants in the probiotic groups[41,47] and from 7 to 152 in the control groups.[42,43] Moreover, different factors, such as the dietary pattern of mothers, mode of delivery, and breastfeeding practice, were not investigated for their effect on the milk microbiome as covariates because of the lack of data from the included studies. Of note, although breastfeeding was mentioned in most studies, whether the infant was breastfed by the mother directly or through the use of pumped milk was not specified. Given the biological difference in these 2 methods, it might serve as another confounding factor affecting the clinical outcomes. Besides, we also pooled the effects of various probiotic species (*Streptococcus thermophilus, B. breve, B. longum, B. infantis, L. acidophilus, L. plantarum, L. paracasei, L. delbrueckii* ssp. *bulgaricus*) in some articles and in others with a limited number of species.[44,55,60] Furthermore, because no review has been conducted so far on this research idea, we have discussed some of our study findings in relation to primary studies.

Conclusion and implications

Overall, oral supplementation with probiotics effectively improved the milk microbiome as evidenced by a higher bacterial detection rate and enriched bacterial abundance, indicating a potential approach for remodeling the milk microbiome by targeting the maternal gut microbiome. Probiotic products, including *Bifidobacterium, Lactobacillus, Streptococcus thermophilus, and Saccharomyces boulardii*, are safe and feasible approaches to boost the abundance of beneficial bacteria in breast milk and to reduce the level of pathogenic bacteria, mainly *Staphylococcus*. Future products should be based on these bacteria to improve maternal and infant health. Importantly, it improves the neonates' gut microbiome, implying that the milk microbiome is an essential route for mother-to-infant bacterial translocation. Furthermore, probiotic treatment has positive metabolic benefits that regulate weight gain and reduce infantile colic risk. Our data further emphasize the importance of breastfeeding in offspring.

Further primary studies are required to update this meta-analysis. Possible future outcomes should be investigated, including (1) the modulatory effect of perinatal oral probiotics on intrauterine microbiome; (2) the effect of oral probiotic supplementation during pregnancy and the postpartum period on different maternal and neonatal health conditions; (3) the modulatory effect of probiotics on different body regions' microbiome in relation to ethnic, seasonal, and setting differences; (4) consideration of breast milk, maternal stool, infant stool, and amniotic fluid as a source of predictive biomarkers for microbiome-related health issues and general health; and (5) breast milk as a source of microbiome transplantation. ■

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ajogmf.2023.101148.

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